

BUSINESS ANALYTICS FOR DATA-DRIVEN DECISION MAKING

An Internship Report submitted to

Mohan Babu University, Tirupati

In partial fulfilment of the requirements for the award of the degree of

**BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING**

Submitted by

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B.Tech. CSE 7th Semester

Under the Esteemed Supervision of

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Department of Computer Science and Engineering

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Sree Sainath Nagar, Tirupati – 517 102

2026 – 2027



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Certificate

This is to certify that the internship report titled "***BUSINESS ANALYTICS FOR DATA-DRIVEN DECISION MAKING***" is the bonafide work done by ***Katuri Yokesh*** (Reg. No: 23102A041000) in the Department of **Computer Science and Engineering**, and submitted to **Mohan Babu University**, Tirupati in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering** during the academic year 2026-2027.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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Dear KATURI YOKESH,

This is to certify that you worked as an Intern in our company from
06/07/2026 TO 02/08/2026.

Please find the internship details below:

Company Name: InternPe

Domain: BUSINESS ANALYTICS

Designation: Intern

During their working period, we found him/her to be a sincere and dedicated intern with a professional attitude and excellent job knowledge.

We thank him/her for his/her efforts and contribution and wish him/her the best in future endeavors.

Yours Sincerely


(Co-Founder)
InternPe



www.internpe.in

ACKNOWLEDGEMENT

I am extremely thankful to our beloved Chancellor, **Dr. M. Mohan Babu Garu**, for his inspiration and unwavering commitment to academic growth and student development.

I express my sincere gratitude to our Pro-Chancellor, **Sri Vishnu Manchu Garu**, his constant encouragement and visionary leadership in fostering a culture of learning and excellence.

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Katuri Yokesb
23102A041000
B.Tech. CSE 7th Semester



MBU
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Vision

To be a globally respected institution with an innovative and entrepreneurial culture that offers transformative education to advance sustainability and societal good.

Mission

- ❖ Develop industry-focused professionals with a global perspective.
- ❖ Offer academic programs that provide transformative learning experience founded on the spirit of curiosity, innovation, and integrity.
- ❖ Create confluence of research, innovation, and ideation to bring about sustainable and socially relevant enterprises.
- ❖ Uphold high standards of professional ethics leading to harmonious relationship with environment and society.

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- ❖ Impart continued education and research support to working professionals in the computing domain to enhance their expertise in the cutting-edge technologies.
- ❖ Inculcate among the computing engineers of tomorrow with a spirit to solve societal challenges.

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- Nurture research, innovation and entrepreneurial skills among faculty and students to contribute to the needs of industry and society.
- Inculcate professional attitude, ethical and social responsibilities for prospective and promising engineering profession.
- Encourage students to engage in life-long learning by creating awareness of the contemporary developments in Computer Science and Engineering and its emerging areas.

B.Tech. Computer Science and Engineering

PROGRAM EDUCATIONAL OBJECTIVES

After few years of graduation, the graduates of B.Tech. CSE will be:

- PEO1.** Pursuing higher studies in core, specialized or allied areas of Computer Science, or Management.
- PEO2.** Employed in reputed Computer and I.T organizations or Government to have a globally competent professional career in Computer Science and Engineering domain or be successful Entrepreneurs.
- PEO3.** Able to demonstrate effective communication, engage in teamwork, exhibit leadership skills and ethical attitude, and achieve professional advancement through continuing education.

PROGRAM OUTCOMES

On successful completion of the Program, the graduates of B.Tech. CSE Program will be able to:

- PO1. Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2. Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions

using first principles of mathematics, natural sciences, and engineering sciences.

- P03. Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- P04. Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- P05. Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- P06. The Engineer and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- P07. Environment and Sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- P08. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- P09. Individual and Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- P010. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- P011. Project Management and Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- P012. Life-long Learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

On successful completion of the Program, the graduates of B. Tech. (CSE) program will be able to:

- PSO1.** Apply knowledge of computer science engineering, Use modern tools, techniques and technologies for efficient design and development of computer-based systems for complex engineering problems.
- PSO2.** Design and deploy networked systems using standards and principles, evaluate security measures for complex networks, apply procedures and tools to solve networking issues.
- PSO3.** Develop intelligent systems by applying adaptive algorithms and methodologies for solving problems from inter-disciplinary domains.
- PSO4.** Apply suitable models, tools and techniques to perform data analytics for effective decision making.

Course Code	Course Title	L	T	P	S	C
22CS111001	Internship	-	-	-	-	2

COURSE OUTCOMES: After successful completion of this course, the students will be able to:

- CO1.** Analyse latest equipment, materials and technologies that are used in industry to solve complex engineering problems following relevant standards, codes, policies and regulations.
- CO2.** Analyse safety, health, societal, environmental, sustainability, economical and managerial factors considered in industry in solving complex engineering problems.
- CO3.** Perform individually or in a team besides communicating effectively in written, oral and graphical forms on practicing engineering.

CO-PO-PSO Mapping Table:

Course Outcome	Program Outcomes												Program Specific Outcomes			
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3		3	3			3				3				
CO2		3				3	3				3					
CO3									3	3						

Correlation Level: 3-High; 2-Medium; 1-Low

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ABSTRACT

This report presents a detailed account of the four-week Business Analytics internship undertaken at InternPe, a technology-driven skill development and internship platform. The internship was designed to build practical competence in business analytics through hands-on work in Microsoft Excel, covering the complete analytics lifecycle from data collection and cleaning to visualization, forecasting, and strategic recommendation. During Week 1, the fundamentals of Business Analytics along with core Excel skills such as sorting, filtering, conditional formatting, data validation, and lookup functions (VLOOKUP, XLOOKUP) were studied and applied to a sales dataset to compute KPIs across e-commerce, retail, and finance business contexts. Week 2 focused on data cleaning and visualization, including customer segmentation using the RFM (Recency, Frequency, Monetary) framework, construction of an interactive sales dashboard, and a SWOT analysis of a simulated startup, TechNova Electronics. Week 3 advanced into trend analysis, financial ratio analysis, break-even computation, linear-regression-based sales forecasting with seasonal adjustment, and a competitive market research study benchmarking TechNova against major Indian e-commerce and retail players. Week 4 consolidated all preceding skills into a capstone project: a complete e-commerce sales analytics dashboard built on a 500-transaction dataset spanning 8 product categories, 5 regions, and 5 sales channels, culminating in data-driven business recommendations across growth, cost reduction, sales improvement, and risk management. The internship strengthened the intern's proficiency in Excel-based data analysis, dashboard design, statistical forecasting, and strategic business communication, while reinforcing the broader value of data-driven decision-making in modern organizations.

List of Acronyms

Acronym	Definition
BA	Business Analytics
KPI	Key Performance Indicator
SWOT	Strengths, Weaknesses, Opportunities, Threats
RFM	Recency, Frequency, Monetary (customer segmentation model)
YoY	Year-on-Year
COGS	Cost of Goods Sold
EBIT	Earnings Before Interest and Tax
OLS	Ordinary Least Squares (regression method)
RMSE	Root Mean Square Error
AOV	Average Order Value
CLV	Customer Lifetime Value
CAC	Customer Acquisition Cost
ROI	Return on Investment
NPS	Net Promoter Score
MRR	Monthly Recurring Revenue
EMI	Equated Monthly Instalment
BNPL	Buy Now, Pay Later
B2B	Business-to-Business
GST	Goods and Services Tax
CRM	Customer Relationship Management

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CHAPTER 1

INTRODUCTION

1.1 Description of the Company

InternPe is a technology-driven internship and skill-development platform that provides students and early-career learners with structured, project-based virtual internship programmes across in-demand domains such as Business Analytics, Web Development, Data Science, Android Development, Machine Learning, and UI/UX Design. The platform is designed to bridge the gap between academic learning and industry-relevant practical skills by assigning interns real-world, task-based projects that simulate the kind of analytical and technical work performed in the industry.

InternPe operates as an MSME-registered and ISO 9001:2015 certified organisation, and its internship programmes are recognised in alignment with AICTE guidelines for skill-based learning. Every internship completed through the platform is validated with a digitally verifiable completion certificate carrying a unique Candidate ID (CID) and a QR-code-based verification link, ensuring authenticity and transparency for academic and recruitment purposes.

The Business Analytics internship track, in particular, is structured as a progressive, four-week curriculum that takes the intern from foundational spreadsheet and analytics concepts through to advanced forecasting and a full capstone analytics project. This structure ensures that interns not only learn individual tools and formulas but also understand how data analysis fits into real business decision-making. The present report documents the work completed by the author during this internship, structured week by week in Chapter 2, along with a reflection on the skills and learning outcomes gained.

1.2 Overview of the Project

The overall project undertaken during this internship was the application of Business Analytics techniques to simulated but realistic business datasets, using Microsoft Excel as the primary analytical tool. The internship was organised into four progressive weekly modules, each building upon the skills acquired in the previous one, so that by the end of the programme the intern was able to independently execute an end-to-end analytics project — from raw data to a strategic business recommendation.

In Week 1, the focus was on understanding the concept and types of Business Analytics (Descriptive, Predictive, and Prescriptive) and on building core Excel competency: sorting and filtering, conditional formatting, data validation, and formulas such as SUM, AVERAGE, COUNT, IF, VLOOKUP, and XLOOKUP, applied to a sales dataset and a set of departmental Business KPIs. Week 2 extended this into data cleaning and visualization, including customer analysis using the RFM (Recency, Frequency, Monetary) segmentation approach, construction of a multi-panel sales dashboard, and a SWOT analysis of a simulated retail startup, TechNova Electronics.

Week 3 introduced advanced analytics: month-on-month trend and seasonality analysis, financial statement analysis (profit and loss, break-even, and expense-structure analysis), linear-regression-based sales forecasting with seasonal adjustment, and a competitive market research study benchmarking TechNova against five major Indian electronics retailers. Week 4 combined all of these skills into a single capstone project — a complete e-commerce sales analytics dashboard covering 500 simulated transactions across 8 product categories, 5 regions, and 5 sales channels for FY2024 — culminating in a structured set of business recommendations covering growth strategy, cost reduction, sales improvement, and risk management.

Across all four weeks, the guiding objective of the project was consistent: to convert raw, unstructured business data into clear, decision-ready insight using the descriptive, predictive, and prescriptive analytics framework introduced in Week 1 — thereby directly practising, at a small scale, the same analytical workflow used by business analysts in real organisations.

1.3 Technology Stack

The internship was deliberately scoped around a single, industry-standard analytics tool — Microsoft Excel — used in depth rather than many tools used shallowly. The specific components of the technology stack used across the four weeks are summarised below.

- Microsoft Excel (core platform): worksheets, formulas, PivotTable-style summary tables, conditional formatting, data validation, and multi-sheet workbook design.
- Excel formulas and functions: SUM, AVERAGE, COUNT/COUNTA, IF, VLOOKUP, XLOOKUP, and nested logical formulas for KPI computation.
- Data visualization in Excel: Bar charts, Line charts, Pie/Doughnut charts, and combination charts used to build the Week 2 and Week 4 dashboards.

- Statistical and analytical methods: Linear Regression (Ordinary Least Squares) for trend forecasting, seasonal-index adjustment, break-even analysis, and financial ratio analysis (Gross Margin, EBIT Margin, Net Margin, ROI).
- Analytical frameworks: SWOT analysis, RFM customer segmentation, and the Descriptive-Predictive-Prescriptive Analytics framework used to structure findings.
- Python with the openpyxl library: used in the Week 4 capstone project for automated generation and structuring of the multi-sheet Excel report.

This stack was chosen because Excel remains the most widely used business analytics tool across small and medium organisations in India, and proficiency in it — combined with statistical reasoning and framework-driven thinking — provides a strong and immediately usable foundation before progressing to specialised BI tools such as Power BI or Tableau.

Table 1.1: Comparison of Descriptive, Predictive and Prescriptive Analytics

Feature	Descriptive Analytics	Predictive Analytics	Prescriptive Analytics
Main Purpose	Understand past events	Predict future events	Recommend the best actions
Key Question	What happened?	What will happen?	What should we do?
Data Used	Historical data	Historical + current data	Historical, current & predictive data
Techniques	Reports, dashboards, data visualization	Statistical models, machine learning	Optimization algorithms, simulations, AI
Output	Business reports and trends	Forecasts and predictions	Recommendations and action plans
Example (this project)	Monthly sales report (Week 1/2)	Sales forecast for Jan-Mar 2025 (Week 3)	Growth, cost & risk recommendations (Week 4)

Table 1.1: Comparison of the three types of Business Analytics, adapted from Week 1.

CHAPTER 2

SUMMARY OF EXPERIENCE

This chapter presents a week-by-week account of the work carried out during the internship, structured to mirror the four weekly modules of the InternPe Business Analytics programme. For each week, the objective, the tasks performed, the tools and techniques applied, and the key findings are documented, supported by tables and figures drawn directly from the work completed.

2.1 Week 1 — Foundations of Business Analytics and MS Excel

Objective: The objective of the first week was to build a conceptual understanding of Business Analytics and to develop practical fluency in Microsoft Excel as the foundational tool for business data analysis.

Work Completed: The week began with a study of Business Analytics and its three principal categories — Descriptive, Predictive, and Prescriptive Analytics (summarised earlier in Table 1.1) — along with their real-world applications in sales forecasting, customer-churn prediction, and pricing optimisation. This conceptual grounding was then applied practically across four hands-on Excel exercises: sorting and filtering a 12-row sales dataset by category and month; applying conditional formatting and data validation to flag high- and low-performing products; using SUM, AVERAGE, COUNT, IF, VLOOKUP, and XLOOKUP formulas to compute GST-inclusive pricing and category-wise aggregates; and analysing a sales dataset to determine Total Sales, Monthly Revenue, the Highest Selling Product, and Average Profit.

The sales-data analysis showed Total Sales of ₹4,13,200 across 12 orders, with the Laptop as the highest-selling product and an Average Profit per order of ₹6,408. Category-wise totals showed Electronics contributing ₹3,20,000, Furniture ₹59,000, and Accessories ₹34,200 — a pattern visualised in Figure 2.1. In parallel, a set of departmental Business KPIs was prepared for Sales, Marketing, HR, Finance, and IT, computing Revenue, Expenses, Profit, Profit Margin, Revenue per Employee, and Customer Satisfaction for each department (Table 2.1 and Figure 2.2), alongside a reference set of KPIs relevant to E-commerce, Retail, and Finance businesses (e.g., Conversion Rate, Customer Acquisition Cost, Inventory Turnover, Loan Approval Rate).

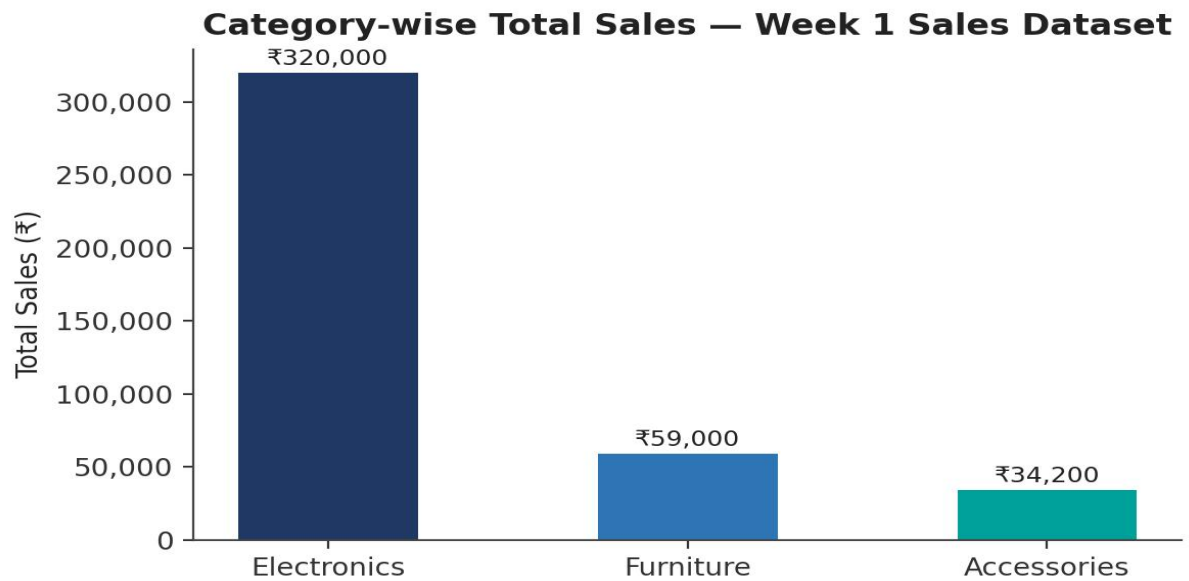


Figure 2.1: Category-wise Total Sales — Week 1 Sales Dataset

Table 2.1: Weekly Business KPI Summary — Week 1

Department	Revenue (₹)	Expenses (₹)	Profit (₹)	Profit Margin	Employees	Cust. Satisfaction
Sales	8,50,000	5,00,000	3,50,000	41.2%	20	88
Marketing	4,20,000	2,50,000	1,70,000	40.5%	12	84
HR	1,80,000	1,20,000	60,000	33.3%	8	90
Finance	3,00,000	1,70,000	1,30,000	43.3%	10	86
IT	5,00,000	3,20,000	1,80,000	36.0%	15	92

Table 2.1: Departmental Business KPIs computed during Week 1.

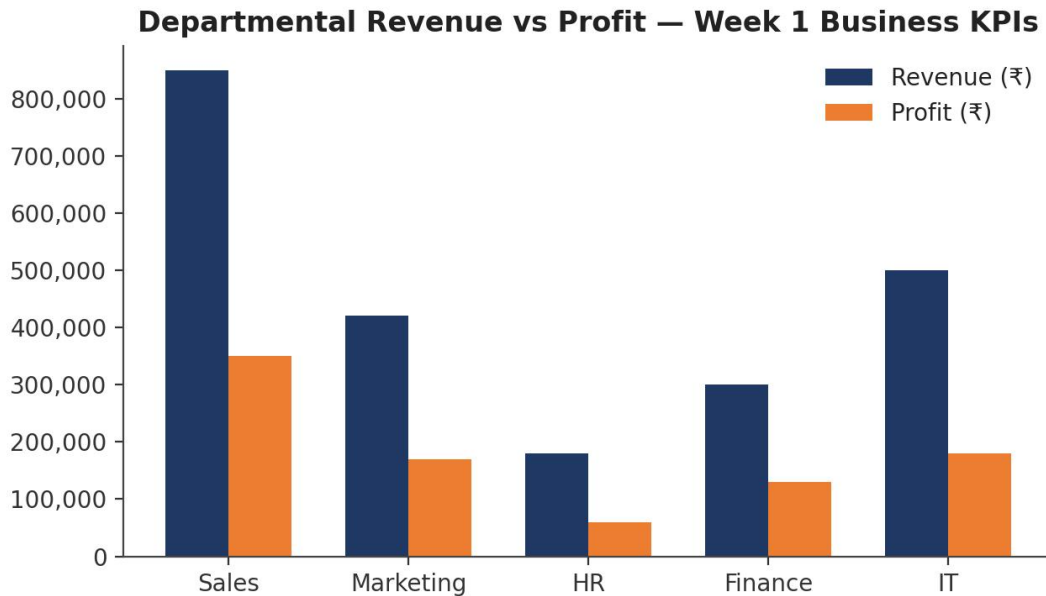


Figure 2.2: Departmental Revenue vs Profit — Week 1 Business KPIs

Skills Gained: Proficiency in Microsoft Excel fundamentals, business data analysis, KPI reporting, and chart-based data visualization, along with an improved conceptual grasp of how descriptive analytics underpins business reporting.

Challenges Faced: Understanding the exact-match behaviour of lookup functions (VLOOKUP/XLOOKUP) and configuring data-validation rules required repeated practice; this was resolved by rebuilding the exercises multiple times with different lookup keys until the logic was fully internalised.

2.2 Week 2 — Data Cleaning, Customer Analytics and Visualization

Objective: The second week focused on data cleaning, exploratory customer analysis, and the design of an interactive, dashboard-style Excel report — the natural next step after building single-formula fluency in Week 1.

Work Completed: A raw transactional dataset was cleaned and standardised (consistent date formats, validated revenue values, and removal of blank/erroneous rows) before being used as the basis for three analytical outputs. First, a Customer Analysis report ranked customers by total revenue and classified them into a four-tier RFM-based segmentation — Premium, Gold, Loyal, and Regular — with the Premium segment (24 of 30 customers, 80% of the base) contributing an average revenue of ₹2,40,546 per customer (Table 2.2 and Figure 2.5). Second, a full Sales Performance Dashboard was built for FY2024, consolidating Total Revenue (₹61,49,440), Total Orders (160), Average Order Value (₹38,434), and the Top

Product (Laptop) into KPI cards, alongside month-wise, region-wise, and product-wise revenue breakdowns (Figures 2.3 and 2.4). Third, a SWOT analysis was carried out for a simulated online electronics retail startup, TechNova Electronics, examining its Strengths (wide product range, strong repeat-customer base), Weaknesses (high dependence on top customers, no physical presence), Opportunities (growing Tier-2/3 smartphone and wearable adoption), and Threats (intense competition from Amazon and Flipkart), together with SO/WO/ST/WT strategic recommendations.

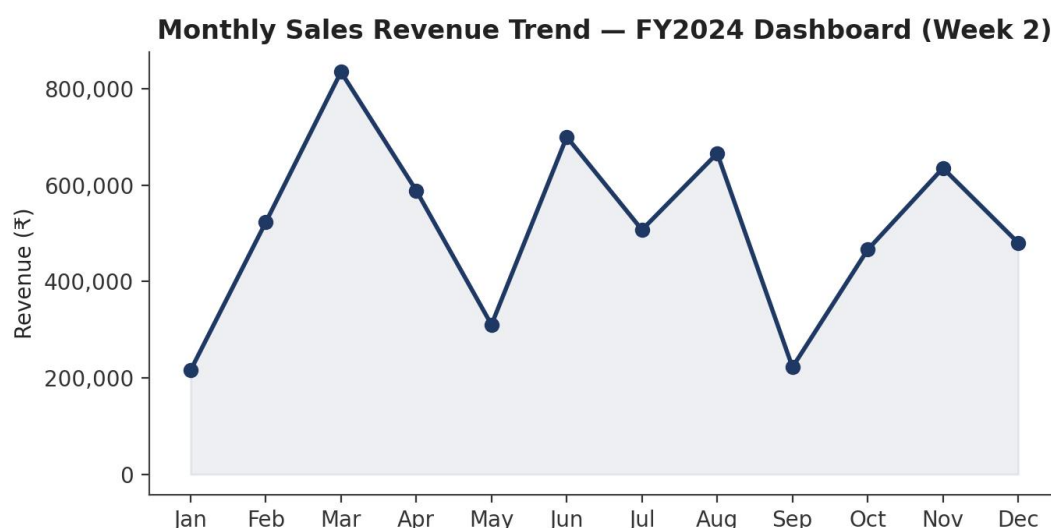


Figure 2.3: Monthly Sales Revenue Trend — FY2024 Dashboard (Week 2)

Region-wise Sales Distribution — Week 2

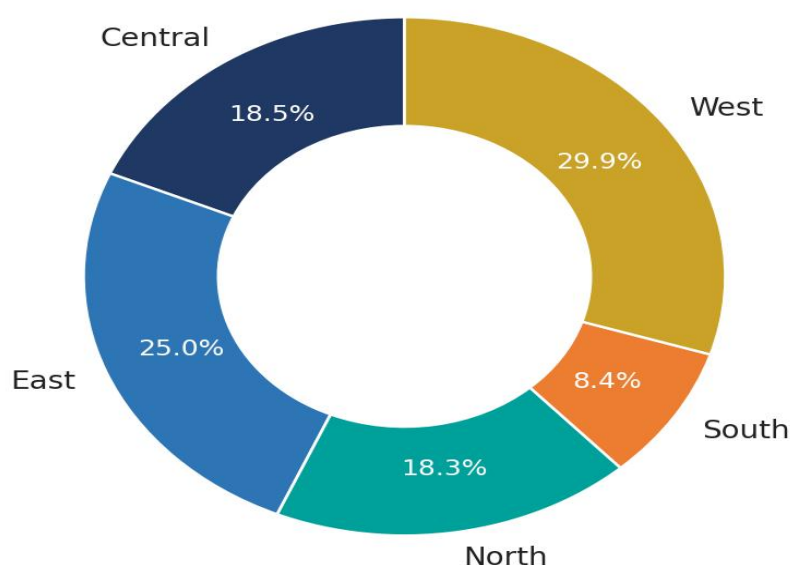


Figure 2.4: Region-wise Sales Distribution — Week 2

Table 2.2: Top 10 Customers by Revenue — Week 2

Rank	Customer	Region	Orders	Revenue (₹)	Segment
1	Manish Agarwal	North	6	4,67,824	Premium
2	Ritu Khanna	East	7	4,20,505	Premium
3	Geeta Mishra	East	5	3,82,441	Premium
4	Anand Bajaj	East	4	3,56,828	Premium
5	Amit Sharma	East	6	3,15,508	Premium

Table 2.2: Top 5 of the Top-10 Customers by Revenue, Week 2 Customer Analysis (abridged).

Customer Segmentation (RFM-Based) — Week 2

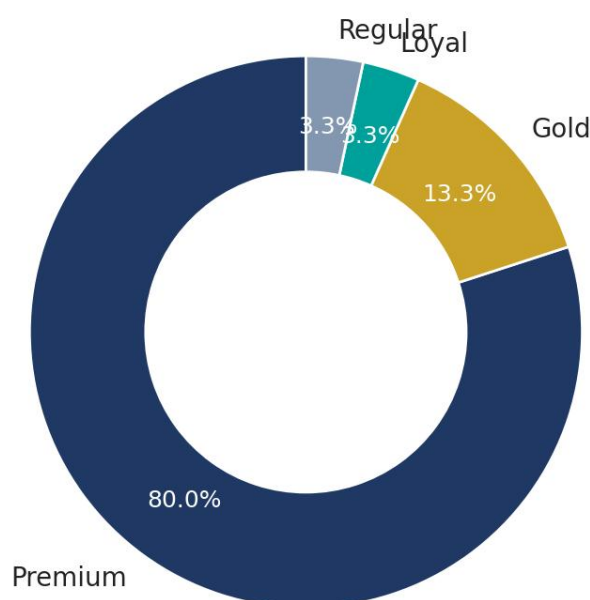


Figure 2.5: Customer Segmentation (RFM-Based) — Week 2

Skills Gained: Data cleaning and validation, customer segmentation using the RFM framework, dashboard design combining KPI cards with multiple chart types, and structured strategic analysis using the SWOT framework.

Challenges Faced: Designing a dashboard that remained readable with many simultaneous data series (monthly, regional, and product-wise) required iterating on chart choice and layout; smaller, focused charts proved more effective than a single crowded visual.

2.3 Week 3 – Advanced Analytics and Forecasting

Objective: The third week moved from descriptive and diagnostic analysis into predictive analytics, introducing statistical forecasting, financial analysis, and competitive market research.

Work Completed: A Trend Analysis compared FY2023 and FY2024 monthly revenue for TechNova Electronics, showing overall growth from ₹66.39 lakh to ₹75.01 lakh (a 13.0% YoY increase), with November identified as the peak month (Diwali-driven demand) and February as the consistent trough (Figure 2.6). A Seasonal Demand Index was computed for each month, showing an index of 1.534 in December (well above the neutral value of 1.0) against 0.684 in February. A Financial Analysis then built a quarterly Profit & Loss statement for FY2024 and computed key financial ratios — Gross Margin (46.0%), EBIT Margin (21.3%), and Net Margin (16.0%) — each benchmarked against an industry target, alongside a formal break-even analysis (Table 2.3).

Building on the trend data, a Sales Forecasting exercise applied an Ordinary Least Squares (OLS) linear regression model to the twelve months of FY2024 revenue, then layered a seasonal-index adjustment on top of the raw trend line to forecast January-March 2025 revenue, with an estimated $\pm 8\%$ confidence band (Figure 2.7). Finally, a Market Research study benchmarked TechNova against five major Indian electronics retailers — Amazon India, Flipkart, Croma, Reliance Digital, and Vijay Sales — across market share, pricing, marketing strategy, and customer sentiment (Figure 2.8), concluding with five prioritised strategic initiatives (e.g., launching a loyalty programme and a mobile app) mapped to expected outcomes and timelines.

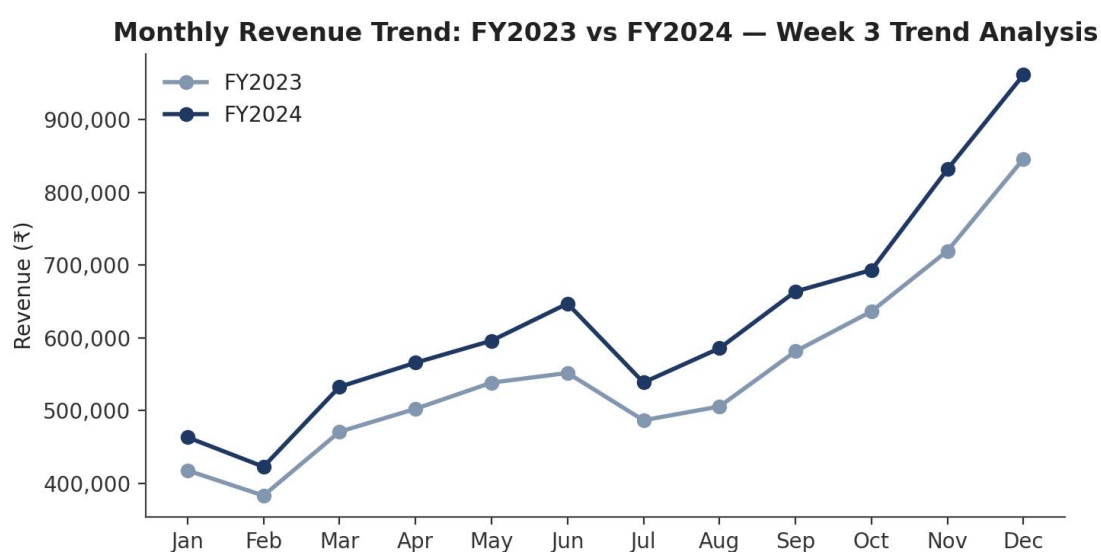


Figure 2.6: Monthly Revenue Trend, FY2023 vs FY2024 — Week 3 Trend Analysis

Table 2.3: Key Financial Ratios — FY2024 (Week 3)

Ratio	Value	Benchmark	Status
Gross Margin	46.0%	40%	Good — above target
EBIT Margin	21.3%	15%	Good — healthy operating efficiency
Net Margin	16.0%	10%	Good — above industry average
COGS Ratio	54.0%	60%	Watch — well controlled
YoY Revenue Growth	13.0%	10%	Good — strong growth

Table 2.3: Key financial ratios computed in the Week 3 Financial Analysis.

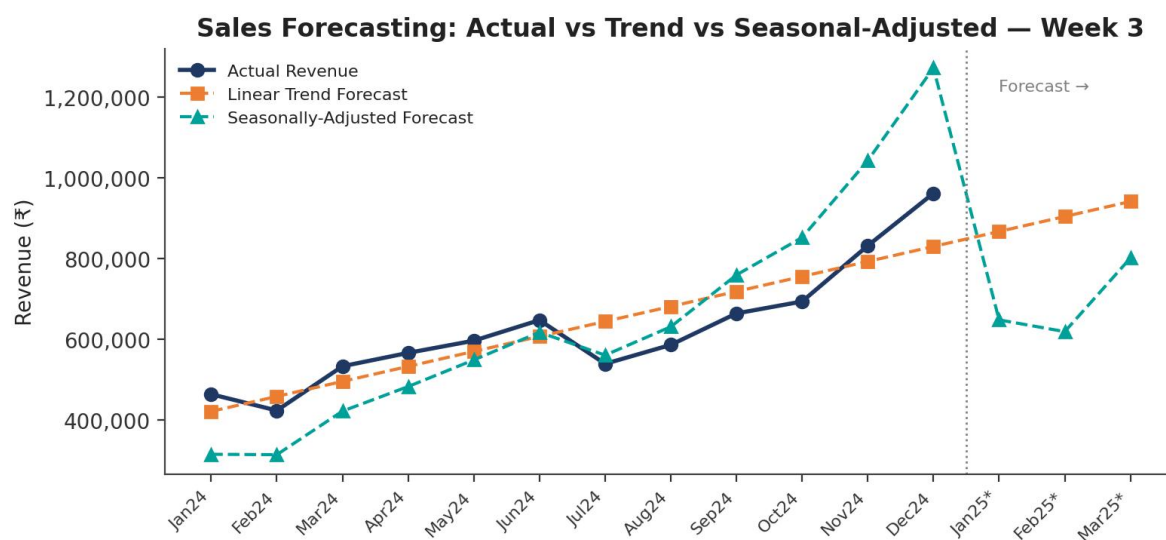


Figure 2.7: Sales Forecasting — Actual vs Trend vs Seasonally-Adjusted Forecast

Indian Electronics Retail — Market Share Benchmark (Week 3)

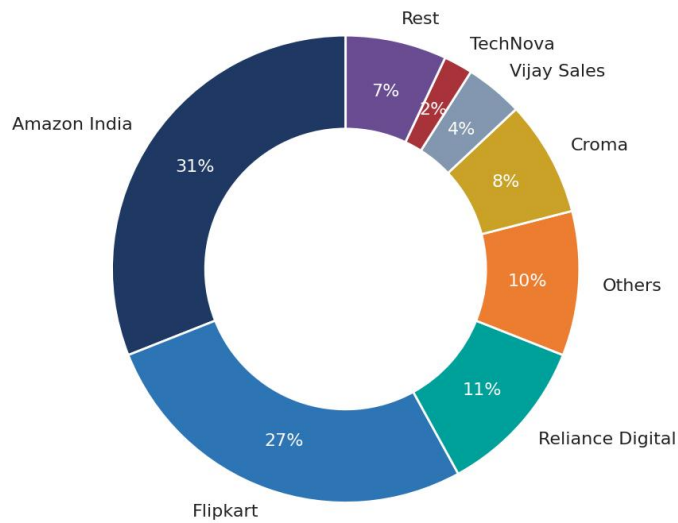


Figure 2.8: Indian Electronics Retail — Market Share Benchmark

Skills Gained: Trend and seasonality analysis, financial-ratio and break-even analysis, linear-regression-based forecasting, and structured competitive/market research.

Challenges Faced: Interpreting the seasonal index correctly (as a multiplier relative to an average month, rather than an absolute value) was initially confusing; working through worked examples for each month resolved this.

2.4 Week 4 — Final Project: E-Commerce Sales Analytics

Dashboard

Objective: The final week consolidated every skill built across the previous three weeks into a single, complete, capstone analytics project — an end-to-end e-commerce sales analytics dashboard.

Work Completed: The project analysed 500 simulated e-commerce transactions from FY2024, spanning 8 product categories, 5 regions, and 5 sales channels, with 22 data fields including revenue, cost of goods sold (COGS), region, channel, customer segment, and product ratings. The methodology followed five structured steps: (1) data collection and synthetic dataset design, (2) data cleaning — standardising dates, validating the revenue-COGS relationship, and removing outliers, (3) exploratory analysis — aggregating the cleaned data by month, category, region, and channel (Table 2.4 and Table 2.5), (4) dashboard development — building a KPI-card-driven Excel dashboard with line, bar, and pie charts, and (5) translating the findings into business recommendations.

The full-year results showed Total Revenue of ₹6,52,93,207 with a Net Profit of ₹3,07,17,810 (a Net Margin of 47.0%) across the 500 orders, an Average Order Value of ₹1,30,586, and a Gross Margin of 47.3% — 7.3 percentage points above the FY25 target of 40% (Figure 2.9). Electronics and Furniture were the two leading categories, together contributing roughly 68% of total revenue, while the Central region generated the highest regional revenue at ₹1,60,34,046 despite having fewer orders than some other regions — indicating a notably higher average order value there (Figures 2.10 and 2.11).

These findings were then translated into a structured set of Business Recommendations across four areas: Growth Strategies (e.g., expanding the mobile-app channel and launching a tiered loyalty programme), Cost Reduction (optimising shipping costs and reducing the return rate from 8.2% toward a 4% target), Sales Improvement (personalised recommendations and discount-strategy optimisation), and Risk Analysis (flagging revenue concentration in the top 3 categories as a high-risk item), together with ten KPIs to monitor against FY25 targets (Table 2.6 and Figure 2.12).

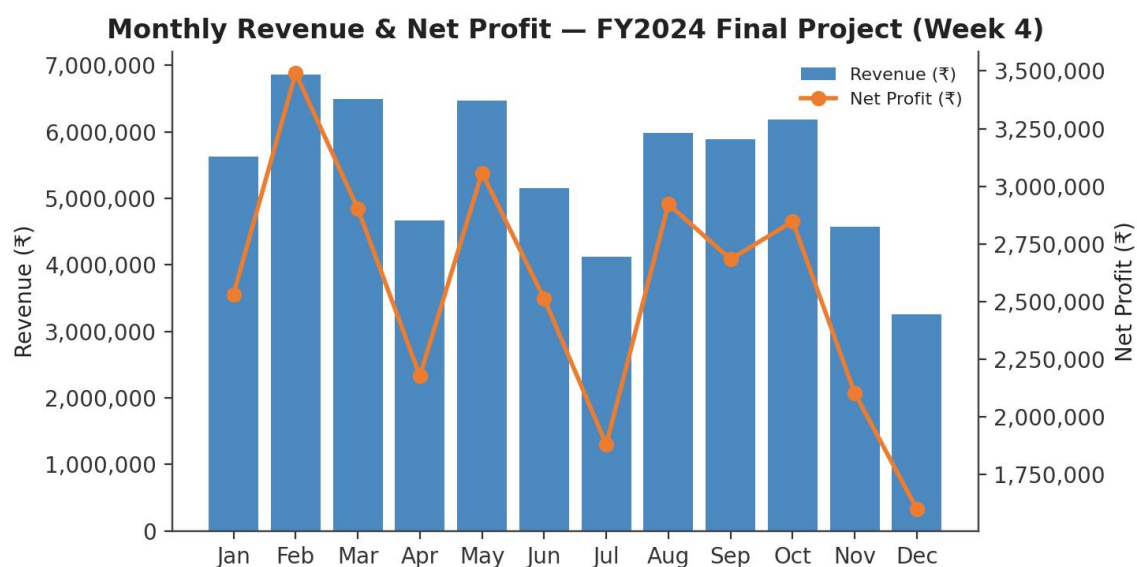


Figure 2.9: Monthly Revenue & Net Profit — FY2024 Final Project

Table 2.4: Monthly Revenue & Profit Summary — FY2024 (abridged, Week 4)

Month	Revenue (₹)	Gross Profit (₹)	Net Profit (₹)	Gross Margin
Jan	56,33,519	25,45,669	25,32,427	45.2%
Feb	68,67,042	35,03,486	34,90,315	51.0%
Jun	51,56,982	25,23,690	25,13,657	48.9%
Oct	61,81,490	28,57,828	28,47,830	46.2%
Dec	32,51,017	16,11,936	16,01,155	49.6%
TOTAL (FY24)	6,52,93,207	3,08,58,268	3,07,17,810	47.3%

Table 2.4: Monthly revenue and profit summary, selected months, FY2024 (Week 4).

Table 2.5: Category Performance Summary — FY2024 (Week 4)

Category	Revenue (₹)	Units Sold	Net Profit (₹)	Margin %
Electronics	2,25,10,890	684	1,04,63,243	46.5%
Furniture	2,19,19,720	824	1,04,77,380	47.8%
Sports	1,06,84,188	610	50,73,250	47.5%
Home & Garden	59,29,856	812	28,09,897	47.4%
Clothing	18,32,493	792	8,40,156	45.8%

Table 2.5: Category-wise revenue, units sold, and margin, FY2024 (Week 4).

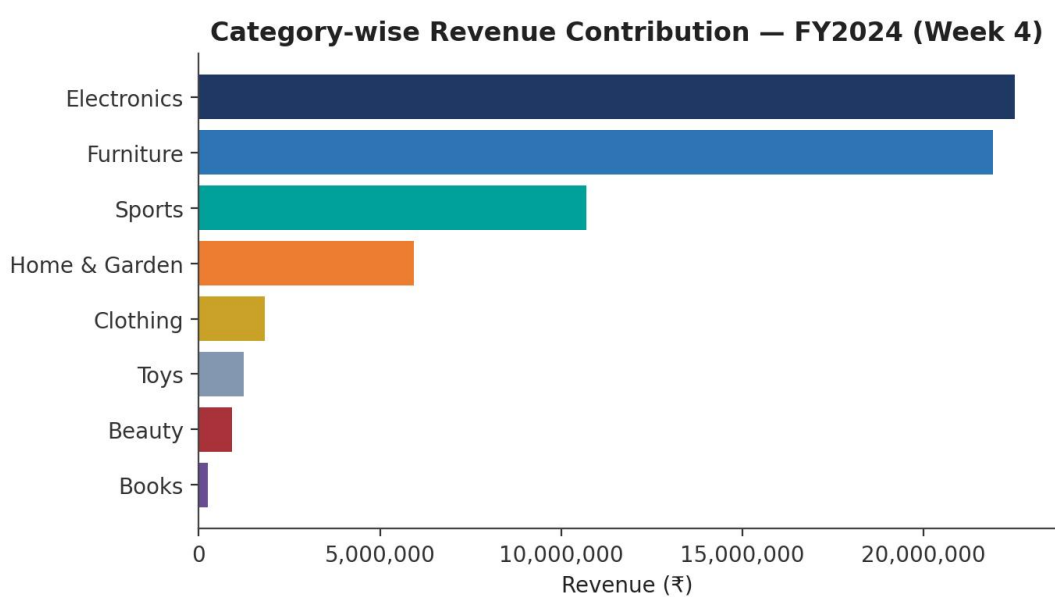


Figure 2.10: Category-wise Revenue Contribution — FY2024

Regional Revenue Contribution — FY2024 (Week 4)

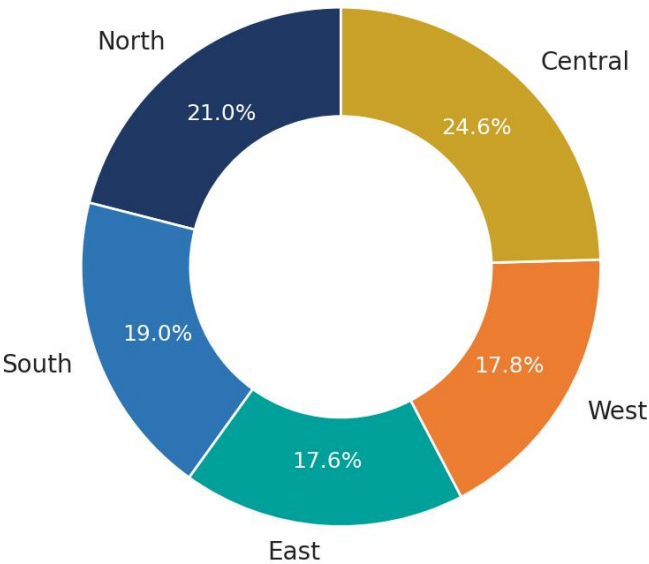


Figure 2.11: Regional Revenue Contribution — FY2024

Table 2.6: Key KPI Targets for FY25 (Week 4)

KPI Metric	Current (FY24)	Target (FY25)	Priority
Gross Margin %	47.3%	≥ 40%	Critical
Customer Retention Rate	67%	≥ 75%	High
Return/Refund Rate	8.2%	≤ 4%	High
Net Promoter Score (NPS)	42	≥ 55	Medium
Mobile App Revenue Share	18%	30%	Medium

Table 2.6: Selected KPI targets set for FY25 based on Week 4 findings.

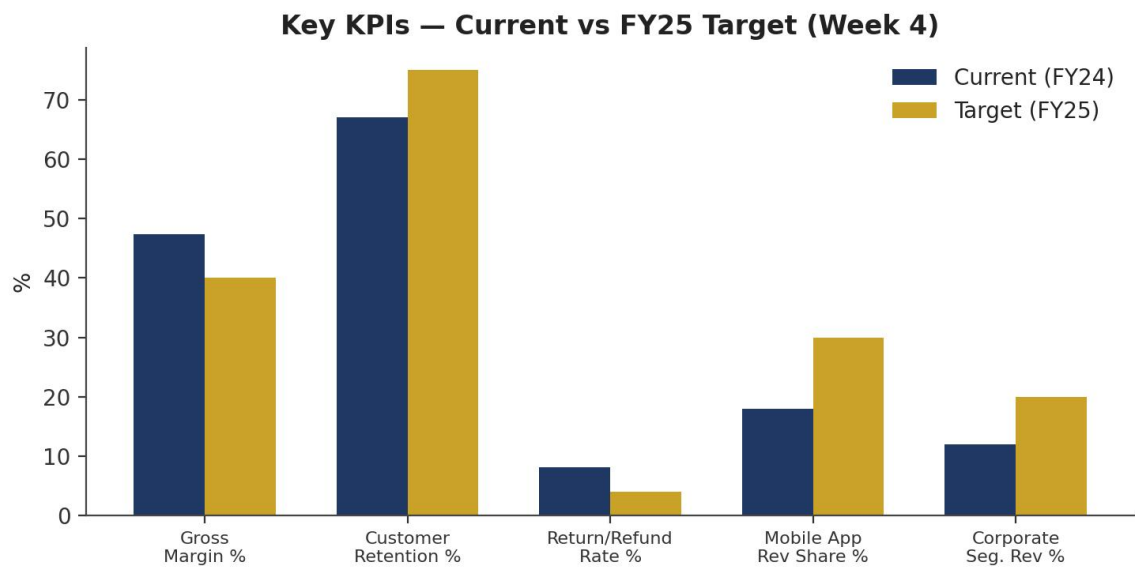


Figure 2.12: Key KPIs — Current vs FY25 Target

Skills Gained: End-to-end analytics project execution — data collection, cleaning, exploratory analysis, dashboard design, and strategic recommendation writing — as well as an introduction to Python (openpyxl) for automated report structuring.

Challenges Faced: Working with a much larger dataset (500 rows, 22 fields) than in earlier weeks made it easy to lose track of data integrity; this was addressed by validating totals against multiple independent aggregation paths (e.g., cross-checking category totals against the monthly totals) before finalising the dashboard.

CHAPTER 3

REFLECTION ON LEARNING

This chapter reflects on the technical and professional skills gained over the four weeks of the internship, the challenges encountered, and how the experience maps onto the Programme Outcomes (POs) and Programme Specific Outcomes (PSOs) of the B.Tech. Computer Science and Engineering curriculum.

3.1 Technical Skills Gained

The most significant technical gain from this internship was a structured, end-to-end fluency in Microsoft Excel for business analytics — progressing from basic formulas and formatting in Week 1 to full dashboard design, statistical forecasting, and multi-sheet report automation by Week 4. Specific competencies developed include: sorting, filtering, conditional formatting, and data validation; lookup formulas (VLOOKUP, XLOOKUP) and logical formulas (IF, nested conditions); KPI design and computation across multiple business functions; chart-based data visualization (bar, line, pie, and combination charts); the RFM customer-segmentation framework and SWOT strategic-analysis framework; financial statement and ratio analysis, including break-even computation; Ordinary Least Squares linear regression for time-series forecasting, with seasonal-index adjustment; and a first exposure to Python's openpyxl library for programmatic spreadsheet generation.

3.2 Professional and Soft Skills Gained

Beyond the technical toolset, the internship also strengthened several professional skills. Working through a structured weekly curriculum with defined deliverables improved time-management and self-discipline, since each week's tasks had to be completed, reviewed, and documented independently. Translating raw analytical output into a business-recommendation format (as in the Week 3 market-research report and the Week 4 recommendations) built business-communication and report-writing skills — learning to state a finding, quantify its impact, and propose a specific, time-bound action. The internship also reinforced attention to detail and data-integrity discipline: because every KPI and chart in later weeks depended on the accuracy of the underlying cleaned data, small errors in Week 2 would have propagated through the entire Week 3 and Week 4 analysis, which encouraged a habit of cross-verifying totals before treating any number as final.

3.3 Challenges Faced and How They Were Overcome

Three recurring types of challenge were faced across the internship. The first was conceptual — for example, correctly interpreting statistical outputs such as the seasonal index in Week 3, or the difference between gross margin and net margin in the financial analysis; these were resolved by working through the underlying formulas manually on a small example before trusting the automated Excel calculation. The second was technical — configuring lookup formulas and data-validation rules correctly in Week 1, and designing a dashboard layout that stayed readable with several simultaneous data series in Week 2 and Week 4; these were resolved through iteration, rebuilding the exercise more than once with different test cases until the result was reliable. The third was scale-related — the Week 4 capstone project used a dataset roughly forty times larger than the Week 1 exercises, which made manual error-checking impractical; this was addressed by building independent cross-checks (e.g., verifying that category-wise revenue totals matched the sum of monthly revenue totals) rather than relying on a single calculation path.

3.4 Mapping to Program Outcomes (POs/PSOs)

The internship activities map naturally onto several Programme Outcomes of the B.Tech. CSE curriculum. The data-cleaning, KPI computation, and forecasting work across all four weeks reflects PO1 (Engineering Knowledge) and PO4 (Conduct Investigations of Complex Problems), since each task required applying quantitative and statistical fundamentals to reach a substantiated conclusion from raw data. The choice and application of Excel formulas, charts, and the OLS forecasting model reflects PO5 (Modern Tool Usage). The written business-recommendation reports produced in Weeks 3 and 4 reflect PO10 (Communication), while independently completing and self-documenting each week's deliverables reflects PO12 (Life-long Learning). Most directly, the entire internship aligns with PSO4 of the CSE programme — the ability to apply suitable models, tools, and techniques to perform data analytics for effective decision-making — which was the explicit and sustained focus of all four weeks of this internship.

CONCLUSION

The four-week Business Analytics internship at InternPe provided a structured, progressively challenging introduction to the practice of business analytics using Microsoft Excel. Starting from foundational spreadsheet skills and the Descriptive-Predictive-Prescriptive analytics framework in Week 1, the internship built steadily through data cleaning, customer segmentation, and dashboard design in Week 2; advanced trend analysis, financial analysis, and statistical forecasting in Week 3; and culminated in a complete, independent capstone analytics project in Week 4 — a 500-transaction e-commerce sales dashboard that produced concrete, quantified business recommendations across growth, cost, sales, and risk.

Across all four weeks, the recurring theme was the conversion of raw data into decision-ready insight: every exercise, from a twelve-row sales sheet in Week 1 to a full-year, 500-order dataset in Week 4, followed the same underlying discipline of cleaning data, computing meaningful KPIs, visualising patterns clearly, and translating the result into a specific, actionable recommendation. This reinforced, in a very concrete and hands-on way, the broader principle that data-driven decision-making — supported by clear KPIs, honest forecasting, and well-communicated recommendations — is central to how modern organisations, regardless of size or sector, plan and improve their performance.

On a personal level, the internship strengthened not only technical proficiency in Excel and statistical/analytical reasoning, but also professional skills in structured self-directed work, business communication, and data-integrity discipline. These skills — proficiency in a widely-used analytics tool, a working knowledge of forecasting and financial analysis, and the ability to communicate findings as business recommendations — are directly transferable to both further academic work and future roles in business analytics, data analysis, or related technology functions, and form a solid foundation on which more advanced tools such as Power BI, Tableau, SQL, and Python for data analysis can be built.

FUTURE SCOPE

While the four-week internship covered the complete analytics lifecycle — data cleaning, visualization, forecasting, and recommendation — it was intentionally scoped around a single tool (Microsoft Excel) and simulated datasets of limited size. Several natural extensions could deepen and scale this work if pursued further.

1. Migration to Business Intelligence Platforms

The dashboards built in Excel during Weeks 2 and 4 could be rebuilt in Power BI or Tableau to support live data connections, drill-down interactivity, and automatic refresh — capabilities that static Excel dashboards cannot easily provide at scale.

2. Database Integration and Automation

The current workflow depends on manually maintained spreadsheets. Storing the transactional data in a relational database (e.g., MySQL or PostgreSQL) and querying it with SQL would allow the same KPI and dashboard logic to run on much larger, continuously updated datasets, and would remove the risk of manual data-entry errors highlighted in Week 2.

3. Advanced and Machine-Learning-Based Forecasting

The Week 3 sales forecast used a linear trend model with a simple seasonal adjustment. Future work could apply more robust time-series methods — such as ARIMA, Holt-Winters exponential smoothing, or Facebook Prophet — and compare their accuracy (using metrics such as RMSE and MAPE) against the simpler model used here, and could extend forecasting to individual product categories and regions rather than only total revenue.

4. Deeper Customer and Product Analytics

The RFM segmentation in Week 2 and the customer analysis in the capstone project could be extended with cohort analysis (tracking how customer groups behave over time), churn prediction using classification models, and market-basket analysis to identify which products are frequently purchased together — insights that would directly support the cross-selling and retention recommendations made in Chapter 2.

5. Application to Real Organizational Data

All datasets used in this internship were simulated. Applying the same analytical workflow — cleaning, KPI computation, dashboarding, forecasting, and recommendation — to a real organization's live sales, marketing, or operations data

would be the most valuable next step, testing whether the skills built here generalize to messier, larger, and less structured real-world data.

Taken together, these extensions represent a natural progression from the Excel-based foundation built during this internship towards the tool-stack (SQL, Python, Power BI/Tableau, and applied statistics) typically used by professional business analysts.

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APPENDICES

Appendix A: Sample of the Raw Transaction Dataset (Week 4 Capstone Project)

The table below shows a representative sample of 8 rows from the 500-row raw transaction dataset used in the Week 4 capstone project, showing 9 of the original 22 fields for readability. The full dataset also included Sub-Category, Product Name, Customer Segment, Customer Age, COGS, Discount %, Shipping Cost, and Order Status fields.

Order ID	Date	Category	Region	Channel	Units	Revenue (₹)	Net Profit (₹)	Rating
ORD-10001	17/11/2024	Clothing	South	Retail Store	4	11,137.57	6,200.61	3.0
ORD-10002	07/10/2024	Electronics	Central	Marketplace	15	7,25,601.90	2,97,591.27	3.2
ORD-10003	27/06/2024	Clothing	North	Mobile App	20	30,865.00	12,596.91	3.9
ORD-10004	12/11/2024	Beauty	North	Online	10	28,106.89	10,554.53	2.9
ORD-10005	27/04/2024	Books	South	Direct Sales	6	2,432.54	519.76	4.2
ORD-10006	21/01/2024	Sports	West	Mobile App	7	2,04,639.79	77,757.34	3.5
ORD-10007	25/08/2024	Home & Garden	South	Direct Sales	19	99,420.46	49,576.44	3.7
ORD-10008	18/01/2024	Clothing	West	Direct Sales	13	23,134.39	10,333.46	4.2

Table A.1: Sample rows from the raw e-commerce transaction dataset, Week 4.

Appendix B: Excel Tools, Functions and Features Used During the Internship

Category	Tools / Functions / Features Used
Lookup & Reference	VLOOKUP, XLOOKUP, INDEX-MATCH
Logical & Conditional	IF, IFS, AND, OR, nested IF
Aggregation	SUM, SUMIFS, AVERAGE, AVERAGEIFS, COUNT, COUNTA, COUNTIFS
Data Preparation	Sorting, Filtering, Data Validation, Remove Duplicates, Text-to-Columns, Find & Replace
Formatting & QA	Conditional Formatting, Custom Number Formats, Data Bars, Colour Scales
Analysis Tools	PivotTables, PivotCharts, What-If Analysis (Goal Seek), Trendlines, Slicers
Charts Used	Column, Bar, Line, Pie/Doughnut, Combo (dual-axis), Scatter

Category	Tools / Functions / Features Used
Statistical Methods	Linear Regression (OLS-based trend line), Seasonal Index Adjustment, RMSE
Frameworks Applied	SWOT Analysis, RFM Customer Segmentation, Break-Even Analysis, YoY Growth Analysis
Other	Python (openpyxl) for dataset generation and validation in the Week 4 capstone project

Table B.1: Consolidated list of Excel tools, functions, and analytical frameworks used across all four weeks.

Appendix C: Complete Top 10 Customers by Revenue

(Week 2 Customer Analysis)

Rank	Cust. ID	Customer Name	Region	Orders	Products	Revenue (₹)	Segment
1	CUST021	Manish Agarwal	North	6	5	4,67,823.90	Premium
2	CUST022	Ritu Khanna	East	7	4	4,20,504.90	Premium
3	CUST018	Geeta Mishra	East	5	4	3,82,441.40	Premium
4	CUST028	Anand Bajaj	East	4	3	3,56,827.50	Premium
5	CUST001	Amit Sharma	East	6	4	3,15,508.40	Premium
6	CUST009	Suresh Nair	Central	9	7	2,89,125.85	Premium
7	CUST029	Swati Bhatt	South	4	4	2,86,910.00	Premium
8	CUST019	Arun Chandra	North	4	4	2,79,854.35	Premium
9	CUST003	Rahul Gupta	South	6	5	2,66,260.70	Premium
10	CUST002	Priya Patel	North	7	6	2,60,450.30	Premium

Table C.1: Full Top-10 customer list by revenue, Week 2 Customer Analysis.